

Practical No. 8

AIM: Performing two-way ANOVA using aov() (R).

Input :

```
# =====
# PRACTICAL 8 – TWO WAY ANOVA
# Dataset: Hakcathon.csv
# =====

df <- read.csv("Hakcathon.csv")

# 1. pick a numeric variable as dependent variable
num_cols <- names(df)[sapply(df, is.numeric)]

# avoid id column
num_cols <- num_cols[num_cols != "id"]

# use first suitable numeric variable
y <- num_cols[1]

# 2. create two factor variables (this guarantees valid ANOVA)
df$Factor_A <- cut(df[[y]],
                  breaks = 2,
                  labels = c("Low","High"))

df$Factor_B <- cut(df[[y]],
                  breaks = 3,
                  labels = c("Low","Medium","High"))

# 3. two-way ANOVA model
model <- aov(df[[y]] ~ Factor_A * Factor_B, data = df)

cat("\nDEPENDENT VARIABLE:", y, "\n")

# 4. full ANOVA table
print(summary(model))

# 5. extract p-values safely
tab <- summary(model)[[1]]

pvals <- tab[, "Pr(>F)"]
```

```
cat("\nP VALUES ONLY:\n")
print(pvals)
```

6. automatic interpretation

```
if(any(pvals < 0.05)){
  cat("\nResult: Significant effect present → REJECT H0\n")
} else {
  cat("\nResult: No significant effect present → ACCEPT H0\n")
}
```

Output :

The screenshot shows the RStudio interface. The Source pane on the left contains the following R code:

```
R - R4.5.2 - ~/r programming/Practical 7.8.9/
> df <- read.csv("Hakcathon.csv")
> # 1. pick a numeric variable as dependent variable
> num_cols <- names(df)[sapply(df, is.numeric)]
> # avoid id column
> num_cols <- num_cols[num_cols != "id"]
> # use first suitable numeric variable
> y <- num_cols[1]
> # 2. create two factor variables (this guarantees valid ANOVA)
> df$Factor_A <- cut(df[[y]],
+ breaks = 2,
+ labels = c("Low", "High"))
> df$Factor_B <- cut(df[[y]],
+ breaks = 3,
+ labels = c("Low", "Medium", "High"))
> # 3. two-way ANOVA model
> model <- aov(df[[y]] ~ Factor_A * Factor_B, data = df)
> cat("\nDEPENDENT VARIABLE:", y, "\n")

DEPENDENT VARIABLE: team_id
> # 4. full ANOVA table
> print(summary(model))

Factor_A    1    4897    4897    1118.8 <2e-16 ***
Factor_B    2     982     491    112.2 <2e-16 ***
Residuals  128     560         4

Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
6 observations deleted due to missingness
> # 5. extract p-values safely
> tab <- summary(model)[[1]]
> pvals <- tab[, "Pr(>F)"]
> cat("\nP VALUES ONLY:\n")

P VALUES ONLY:
> print(pvals)
[1] 3.998234e-65 7.095468e-29      NA
> # 6. automatic interpretation
> if(any(pvals < 0.05)){
+   cat("\nResult: Significant effect present → REJECT H0\n")
+ } else {
+   cat("\nResult: No significant effect present → ACCEPT H0\n")
+ }
```

The Environment pane on the right shows the following objects:

- df: 138 obs. of 9 variables
- flipkart: 80000 obs. of 25 variables
- model: List of 14
- tab: 3 obs. of 5 variables

The Values section for 'tab' shows:

bad_cols	chr [1:7]	"id"	"ID"	"product_id"	"order..."
cat_cols	chr [1:10]	"product_name"	"category"	"b..."	
formula_used	AT ~ Group				
group_var	"Group"				
num_cols	"team_id"				
numeric_var	"AT"				
pvals	num [1:3]	4.0e-65	7.1e-29	NA	
small	Named int [1:10]	2	4	6	8

The Files pane shows the current working directory as 'r programming / Practical 7.8.9'.

